

William Jackson

Senior Software Engineer – Vue C# (.NET) Azure/AWS

Wellington, Nevada, United States | +1 (786) 949-7561 | williamjdev@outlook.com | [linkedin.com/in/william-jackson-40036a3ba](https://www.linkedin.com/in/william-jackson-40036a3ba)

PROFESSIONAL SUMMARY

Senior Software Engineer operating within **SaaS, fintech, and data-intensive platforms**, delivering resilient full-stack systems using **Vue.js (2–3), TypeScript (3–5), and C#/.NET / ASP.NET Core (2–8)** across cloud-native environments. Expertise spans **microservices, event-driven architecture, and secure API ecosystems**, consistently improving system reliability by over **40% through observability, idempotency, and backpressure strategies**. Proven in modernizing legacy platforms into scalable distributed systems, implementing **OpenTelemetry-driven monitoring**, and optimizing performance under high concurrency while ensuring secure, maintainable, and production-grade delivery.

SKILLS

- **Frontend** — **Vue.js (2–3), TypeScript (3.8–5.x), JavaScript (ES6+)**, HTML5, CSS3, Tailwind CSS, Bootstrap, component design systems, accessibility, state management, SPA routing, performance profiling
- **Backend & APIs** — **C# (7–12), .NET Framework (4.6–4.8), ASP.NET Core (2.1–8)**, REST APIs, microservices, background jobs, WebSockets, API versioning, rate limiting, idempotency, secure coding (OWASP-aligned)
- **Data, Messaging & Search** — **PostgreSQL (11–16)**, MySQL, SQL Server, **Redis (5–7)**, MongoDB, DynamoDB, **Kafka (2.x–3.x)**, RabbitMQ, AWS SQS/SNS, **Elasticsearch (6–8)**, OpenSearch, Redshift, BigQuery
- **Cloud & DevOps** — **AWS, Docker (20.x–26.x), Kubernetes (1.20–1.30)**, Helm, Argo CD, GitOps, **Terraform (1.0–1.7)**, CloudFormation, GitHub Actions, GitLab CI, Jenkins, Azure DevOps, CircleCI, Azure, GCP
- **Observability, Quality & Security** — **OpenTelemetry, Datadog, Prometheus, Grafana**, ELK/EFK, CloudWatch, Jest, Mocha, Chai, React Testing Library, Cypress, Playwright, Pact-style contract tests, **OAuth2, OpenID Connect, JWT, Okta, Auth0, AWS Cognito, RBAC/ABAC**, audit logging

WORK HISTORY

Senior Software Engineer

February 2021 to December 2025

Viaro Networks Inc. - Delaware, United States

- Delivered scalable **Vue.js (2–3) frontend architecture** with reusable component systems that reduced UI rework by over **35% across multiple product teams** while improving maintainability and consistency.
- Engineered **BFF and microservice layers using C#/.NET (5–9) and ASP.NET Core**, stabilizing frontend-backend contracts and eliminating integration fragility across evolving services.
- Designed robust **REST API ecosystems** with consistent error handling, validation, and pagination, significantly improving frontend predictability and reducing production defects.
- Implemented **event-driven pipelines using Kafka**, resolving duplicate processing issues through idempotent consumers and achieving over **99.9% processing accuracy**.
- Built resilient asynchronous workflows with **AWS SQS/SNS and DLQ strategies**, enabling safe retries and large-scale data backfills without impacting live traffic.
- Optimized **PostgreSQL query performance** using indexing and execution plan tuning, improving API latency by more than **45% on high-load endpoints**.
- Introduced **Redis-based caching layers** and throttling mechanisms, reducing database load during peak traffic conditions by over **30%**, with consistent caching patterns applied across **AWS ElastiCache and Azure Cache for Redis** deployments.
- Delivered high-performance search capabilities using **Elasticsearch and OpenSearch**, enabling real-time filtering and investigation workflows across distributed application data and **Azure-integrated operational services**.
- Developed analytics pipelines integrating **Redshift, BigQuery, and Azure SQL** reporting flows, improving cross-team reporting capabilities and strengthening data-driven decision making.

- Strengthened security architecture using **OAuth2, OpenID Connect, JWT, AWS Cognito, and Azure AD / Microsoft Entra ID**, ensuring consistent enforcement of **RBAC** and **ABAC** policies across platform services and internal tools.
- Standardized distributed tracing with **OpenTelemetry** and built monitoring dashboards in **Datadog, Grafana, CloudWatch, and Azure Application Insights**, improving incident detection speed by over **50%**.
- Automated deployments using **Docker, Kubernetes (EKS/AKS), ECS, Helm, and Terraform**, enabling reliable CI/CD pipelines and repeatable infrastructure provisioning across multi-environment cloud delivery.

Software Engineer
Avantia, Inc. - Ohio, United States

October 2018 to January 2021

- Developed **Vue.js (2.x) operational dashboards** that reduced manual support workflows and improved internal efficiency across operations teams.
- Architected backend services using **ASP.NET Core (.NET Core 2–5)** with clear domain separation, reducing cross-team dependencies and improving scalability.
- Designed RESTful services with stable payload contracts, reducing frontend regressions and improving integration reliability across releases.
- Implemented **RabbitMQ-based background processing systems**, resolving retry failures and ensuring reliable task execution under load.
- Enhanced streaming integrations using **Kafka**, solving consumer lag and poisoned message issues through partition tuning and validation strategies.
- Improved database reliability across **PostgreSQL, MySQL, and SQL Server** by enforcing migration discipline and optimizing slow queries.
- Introduced **Redis caching standards**, improving response times and reducing database contention under heavy workloads.
- Built vendor integration layers that isolated third-party inconsistencies, allowing internal systems to evolve independently.
- Implemented secure authentication using **OAuth2 and JWT**, ensuring consistent authorization enforcement across services and UI layers.
- Enhanced observability through **OpenTelemetry and Grafana dashboards**, significantly improving debugging of intermittent production issues.
- Expanded automated testing coverage using **Jest, Mocha, and Chai**, reducing regression rates during rapid deployment cycles.

Software Developer
ArnAmy, Inc. - Florida, United States

March 2016 to September 2018

- Developed internal tools using **Vue.js (early 2.x)** to visualize workflows and operational data, improving team productivity and transparency.
- Built backend APIs with **C#/.NET Framework transitioning to early .NET Core**, unifying multiple upstream systems into consistent service layers.
- Implemented strict validation and error-handling strategies, preventing unstable UI flows during partial service failures.
- Designed transactional data systems across **PostgreSQL and MySQL**, eliminating partial writes and improving financial reconciliation accuracy.
- Automated reporting workflows with scheduled jobs and retry mechanisms, replacing manual processes with reliable pipelines.
- Improved performance by analyzing slow queries and introducing targeted indexing strategies, reducing reporting delays significantly.
- Built ingestion pipelines with schema validation and quarantine mechanisms, preventing corrupted data from entering production systems.
- Introduced **MongoDB for flexible document storage**, improving handling of semi-structured metadata alongside relational systems.
- Implemented caching with **Redis**, improving read-heavy API performance and stabilizing backend load.

- Delivered search capabilities using **Elasticsearch**, enabling fast querying without overloading transactional databases.

Junior Software Developer

October 2014 to February 2016

Centric Tech Inc. - New Jersey, United States

- Developed internal admin tools using **server-rendered UI and early JavaScript-enhanced SPA patterns**, improving usability of internal workflows.
- Built **REST-style APIs** using **C#/.NET Framework**, standardizing validation and error responses to reduce support overhead.
- Implemented data access layers using **SQL Server**, improving transaction reliability and reducing inconsistencies in business operations.
- Enhanced background job processing with **RabbitMQ**, introducing retry and backoff strategies to improve job stability.
- Added **Redis-based caching** for high-read workflows, reducing redundant database queries and improving application responsiveness.
- Implemented internal search and filtering with **Elasticsearch**, enabling faster lookup for operational users without increasing database load.
- Standardized build and deployment pipelines using **Jenkins**, eliminating environment-specific deployment issues.
- Added targeted unit tests for business rules using **NUnit / MSTest**, reducing regression defects in production.
- Collaborated with senior engineers to diagnose performance bottlenecks using logs and query profiling techniques, improving system efficiency.

EDUCATION

Bachelor's degree in Computer Science

2010 to 2014

Harvard University

- Built strong foundations in **algorithms, databases, and software engineering**, applying coursework through practical team projects and production-style system design.

CERTIFICATIONS

- **AWS Certified Developer – Associate (Preparation & Practical Implementation)**
- **Advanced .NET Backend Engineering & API Design**
- **Modern Frontend Architecture with Vue.js & TypeScript**